DRUGS · PAIN AND NAUSEA

# NSAIDs: ibuprofen and diclofenac

Non-steroidal anti-inflammatory drugs inhibit cyclo-oxygenase and reduce prostaglandin-mediated pain and inflammation. Ibuprofen is the standard partner to paracetamol after tonsillectomy and other ENT surgery, and a systematic review found that NSAIDs (other than ketorolac) do not increase the risk of post-tonsillectomy bleeding requiring a return to theatre. Their harms are on the stomach, the kidney, the heart, and in some patients with asthma.

## 1 CHAPTER ONE At a glance

The drugs on one page.

### 1.1 Summary

NSAIDS

| | SUMMARY |
|---|---|
| **Agents** | Ibuprofen; naproxen; diclofenac; ketorolac (parenteral); celecoxib and etoricoxib (COX-2 selective) |
| **How they work** | Inhibit cyclo-oxygenase 1 and 2, reducing prostaglandins and thromboxane; analgesic, anti-inflammatory and antipyretic |
| **ENT uses** | Post-operative analgesia (tonsillectomy, ear, nasal, thyroid); acute otitis media and sore throat; inflammatory conditions (salivary, TMJ) |
| **Adult dose** | Ibuprofen 400 mg three times daily (maximum 2.4 g a day); naproxen 250 to 500 mg twice daily; diclofenac 50 mg two or three times daily (maximum 150 mg a day) |
| **Child dose** | Ibuprofen 5 to 10 mg/kg three or four times daily (maximum 30 mg/kg a day, up to 2.4 g), from 3 months and 5 kg |
| **Avoid** | Active peptic ulcer or bleeding; severe heart failure; severe renal impairment; aspirin-exacerbated respiratory disease; third trimester of pregnancy; diclofenac in ischaemic heart, peripheral arterial or cerebrovascular disease; dehydration (renal injury); chickenpox (serious skin infection reported) |

# 2 CHAPTER TWO Using them in ENT

Practice.

## 2.1 Tonsillectomy and bleeding

The Cochrane review and subsequent work show that ibuprofen and similar NSAIDs, given with paracetamol, give good analgesia after tonsillectomy without a significant increase in bleeding needing surgery, and reduce the need for opioids, which is particularly important in children with sleep apnoea. Many guidelines (including the AAO-HNS) support ibuprofen after tonsillectomy.

## 2.2 Asthma

Most asthmatics tolerate NSAIDs. Those with **aspirin-exacerbated respiratory disease** (asthma, nasal polyps, and reactions to aspirin or NSAIDs) must avoid COX-1 inhibitors; COX-2 selective agents are usually tolerated.

# 3 CHAPTER THREE Safety

What goes wrong.

## 3.1 Adverse effects

- Dyspepsia, peptic ulceration and **gastrointestinal bleeding**: give a PPI with long-term use or in the elderly.
- **Acute kidney injury**, particularly with dehydration, ACE inhibitors and diuretics (the "triple whammy").
- Fluid retention, hypertension, heart failure, and increased cardiovascular events (diclofenac and COX-2 inhibitors more than naproxen and low-dose ibuprofen).
- Bronchospasm in aspirin-sensitive asthma.

### 3.2 Interactions

Warfarin, DOACs, SSRIs and steroids (bleeding); ACE inhibitors and diuretics (renal failure); methotrexate and lithium (toxicity).

## 4 CHAPTER FOUR For the examination

What the examiner wants to hear.

### 4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The evidence on **NSAIDs and post-tonsillectomy bleeding**.
- **Aspirin-exacerbated respiratory disease** and NSAIDs.
- The triple whammy and other key cautions.

### 4.2 Common pitfalls

COMMON PITFALLS

- Withholding ibuprofen after tonsillectomy and prescribing codeine instead.
- NSAIDs in a dehydrated child who is not drinking.

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